

# Week 8 Report

16 - 20 August 2026 | UFACTORY xArm 6 work cell

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## This Week in Short

The week split in two. The first day went to the arm: starting a fresh sorting cell and, more importantly, finding the reason the previous one could not pick. The rest went to written work - a full report of all six projects, the guide book for the arm, and the training presentation.

## A Fresh Start on the Sorting Cell

The new cell has the camera on a stand off to the side; the arm lifts the blue cubes with a suction cup and leaves the red ones. This is a restart, not a repair. The earlier fixed-camera project has the right design and passes all its own checks, but has never completed a pick, and the reason is one measurement: the link between what the camera sees and where the arm must go is off by about 17 mm, while a cup on a cube allows roughly 6 mm.

## Finding the Real Cause

The important result was tracing that error to its source, and it was not the camera. The old calibration was done by hand: the arm was left loose and I pressed the cup onto a cube twelve times, letting the vacuum sensor record each contact. That sensor confirms the cup sealed but is blind to whether it was centred - a seal happens anywhere within about 11 mm of the cube's middle. A hand also presses to a different depth every time, which showed as nearly 19 mm of variation in the recorded heights, and the wrist angle was free and never written down. Those three account for the error. The replacement removes the hand entirely: stick a cube to the cup - crookedly is fine - and let the arm drive itself through a grid of 27 positions while the camera watches the cube it is holding. The offset never changes and cancels itself out, there is no contact to press unevenly, and the wrist angle is commanded rather than guessed.

## Tools Built

An offline check suite that runs safely with no arm and no camera connected; two measurement checks that report what the camera sees without commanding a single move; a hand-jog tool; and the calibration fit itself, which now refuses to save a result that is not good enough. The previous calibration wrote a poor answer to disk with only a warning, and every tool afterwards trusted it. A repeatability gate also runs first: measure one motionless cube fifty times and check the readings agree. The first run gave 2.31 mm of spread, which the tool flags as marginal, so that is the next thing to understand.

## Documentation

A full report covering all six projects built on this arm, written as plain description with no code: what each was for, how it works, what went wrong, how it was fixed, and where it honestly stands. The xArm 6 guide book was finished and re-issued in the KACST format. From the training report I then built the King Saud University co-op presentation - 24 slides in the university theme, covering the 4-DOF arm from phase one, the six projects of phase two, the measured results, the safety practices, and the lessons and recommendations - with short video clips of the arm working.

## Where Things Stand, and Next

The taught pick-and-place and the vision picker remain the two projects proven on hardware. In the sorting cell everything except one number works: camera, cup, pump, seal sensor, arm motion, detection and both offline suites all pass, and the wrist fault that blocked the vacuum tool is solved. Calibration is the only blocker left, designed but not yet run. Next: run it behind the repeatability gate, then verify it independently with a cube in each corner against a tape measure rather than trusting the fit's own score; then teach the table surface, run detection end to end, and attempt the first stationary pick. The two drop spots for the dynamic picker are still one short teaching session away.